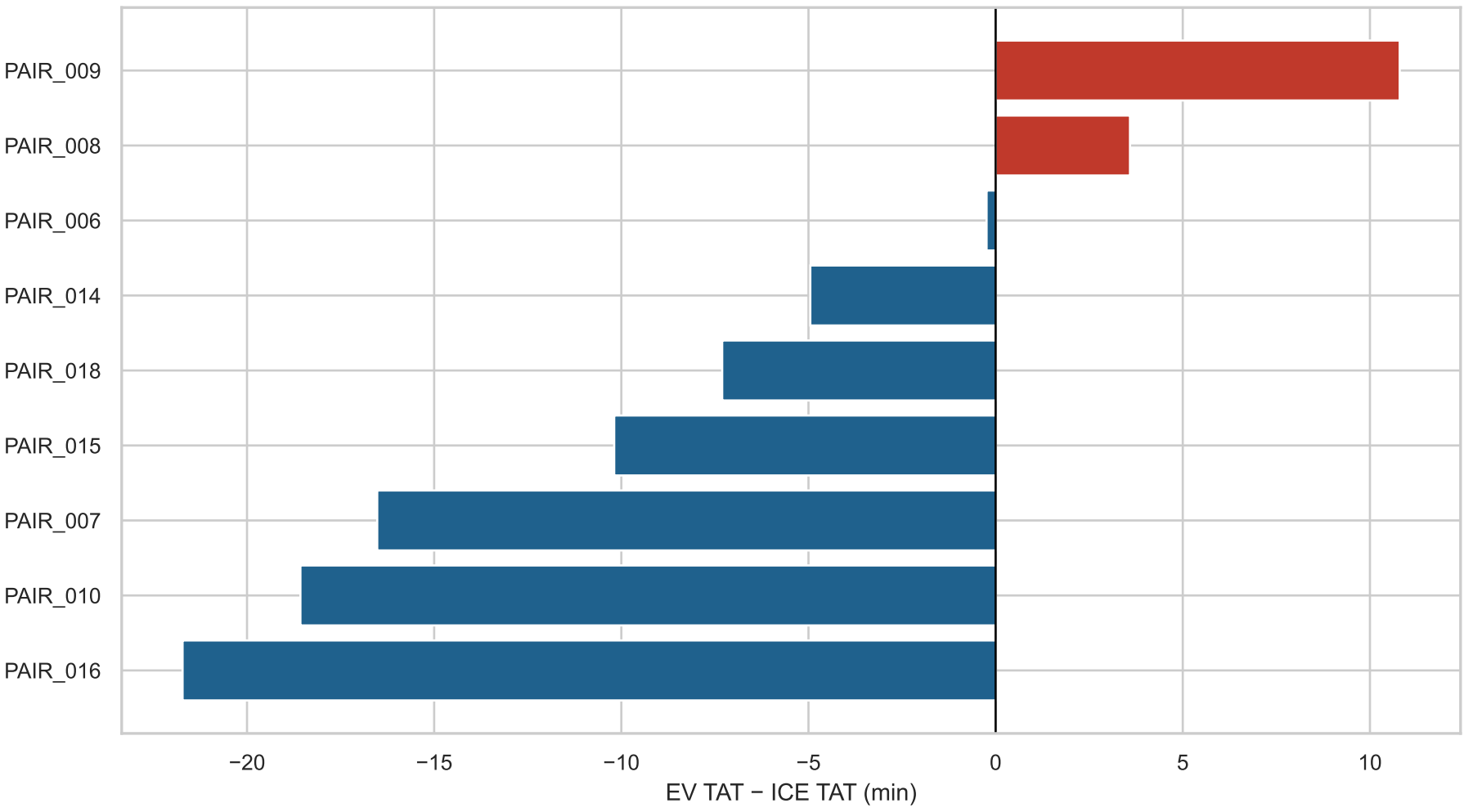
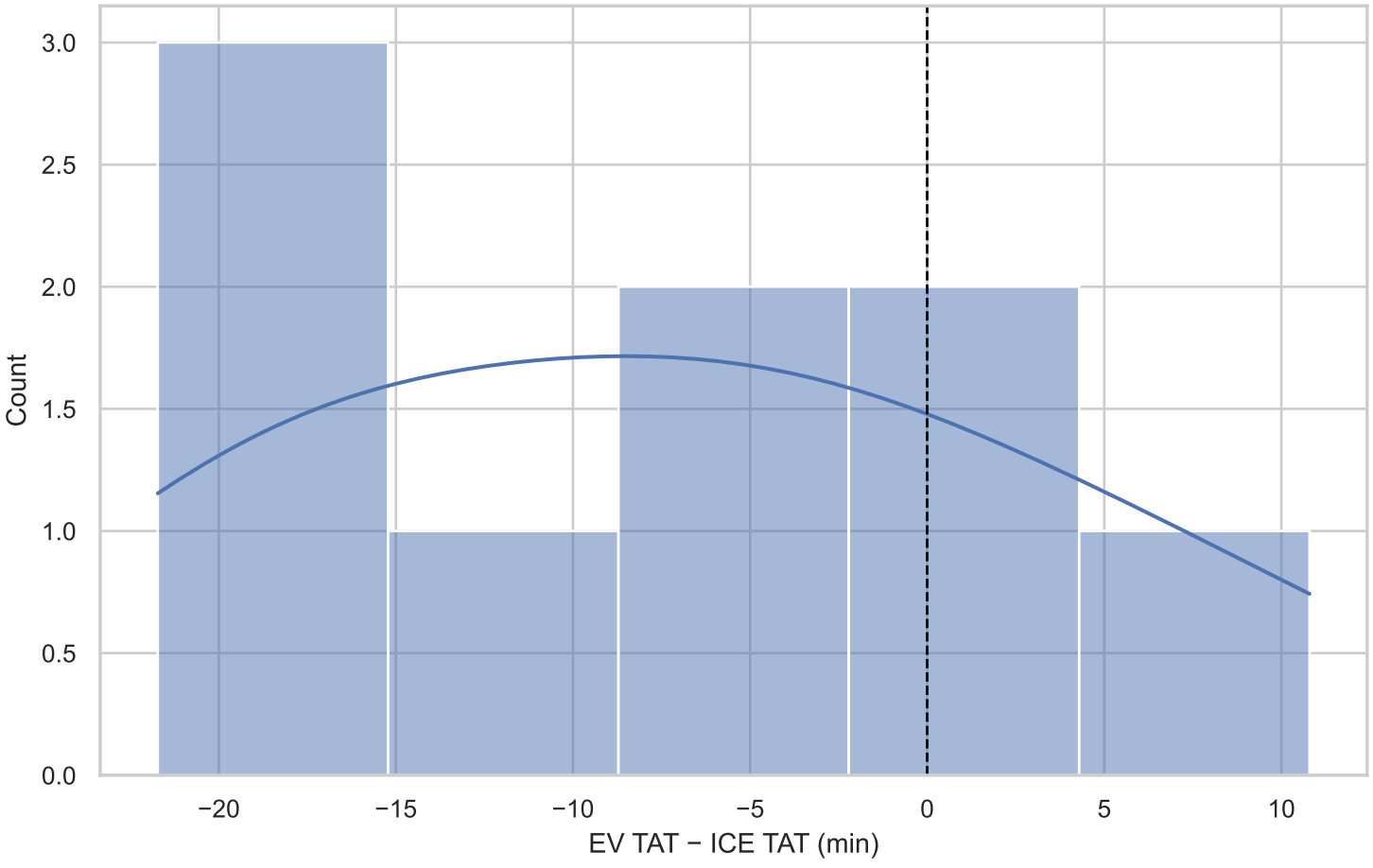
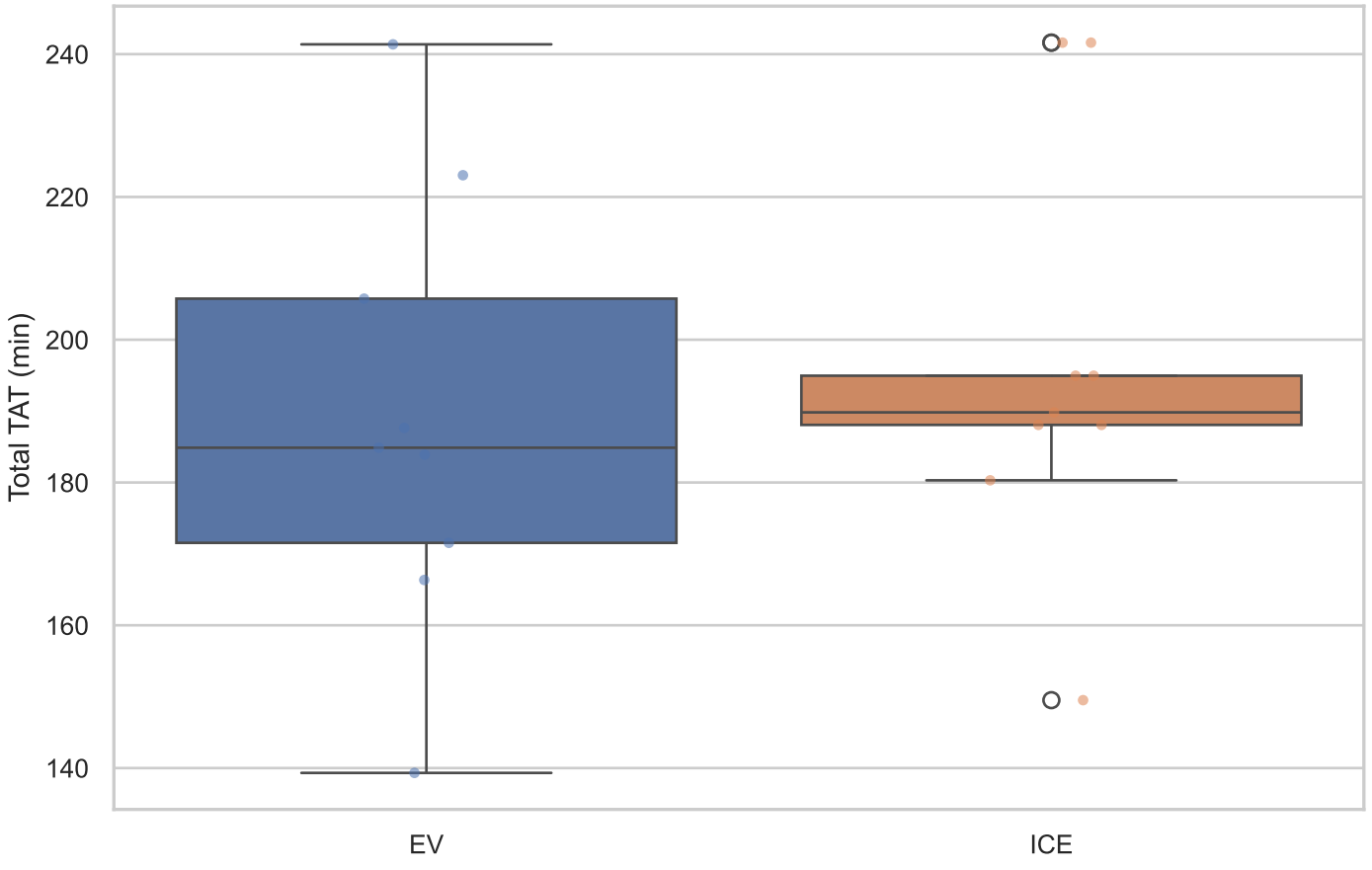
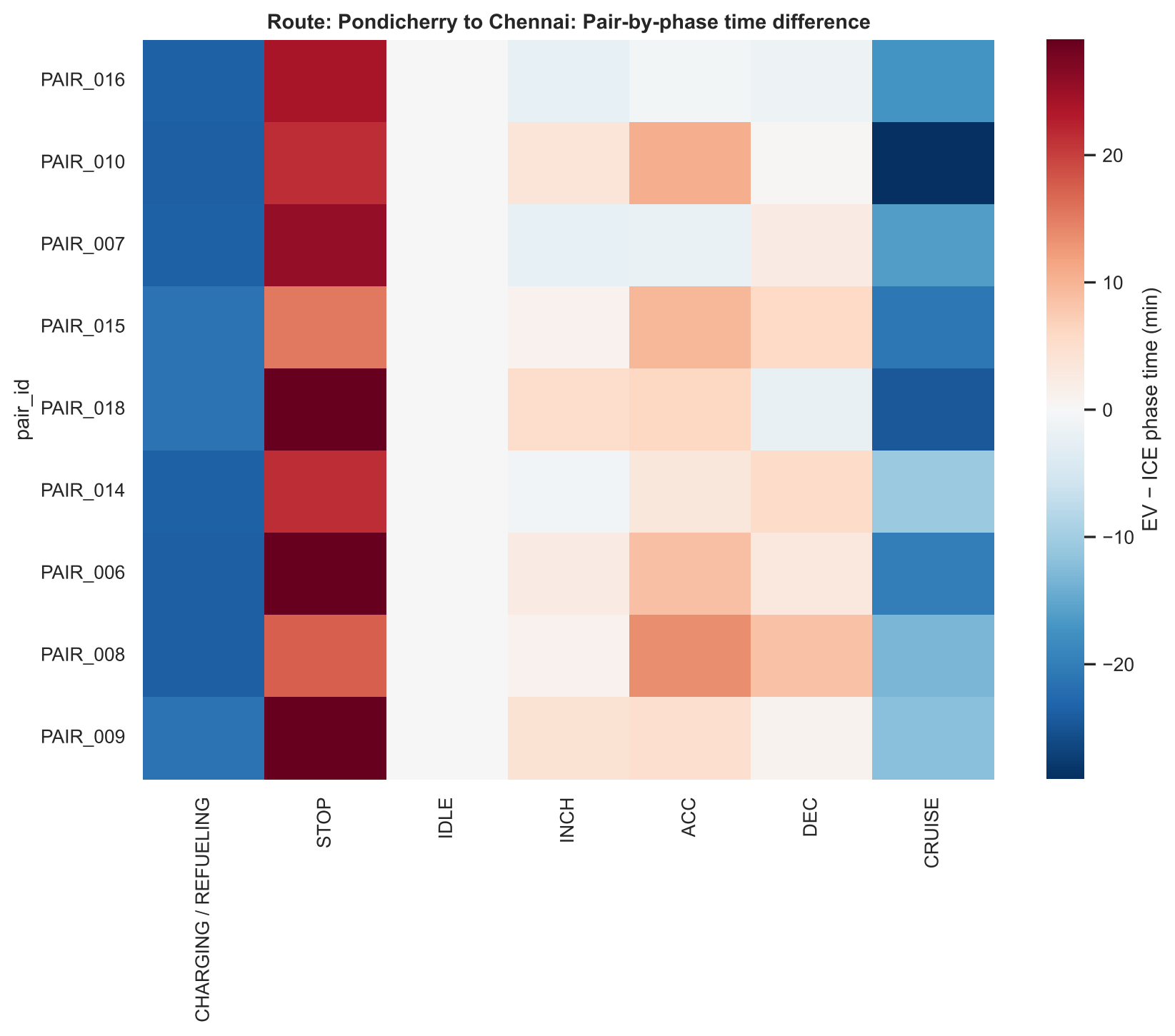
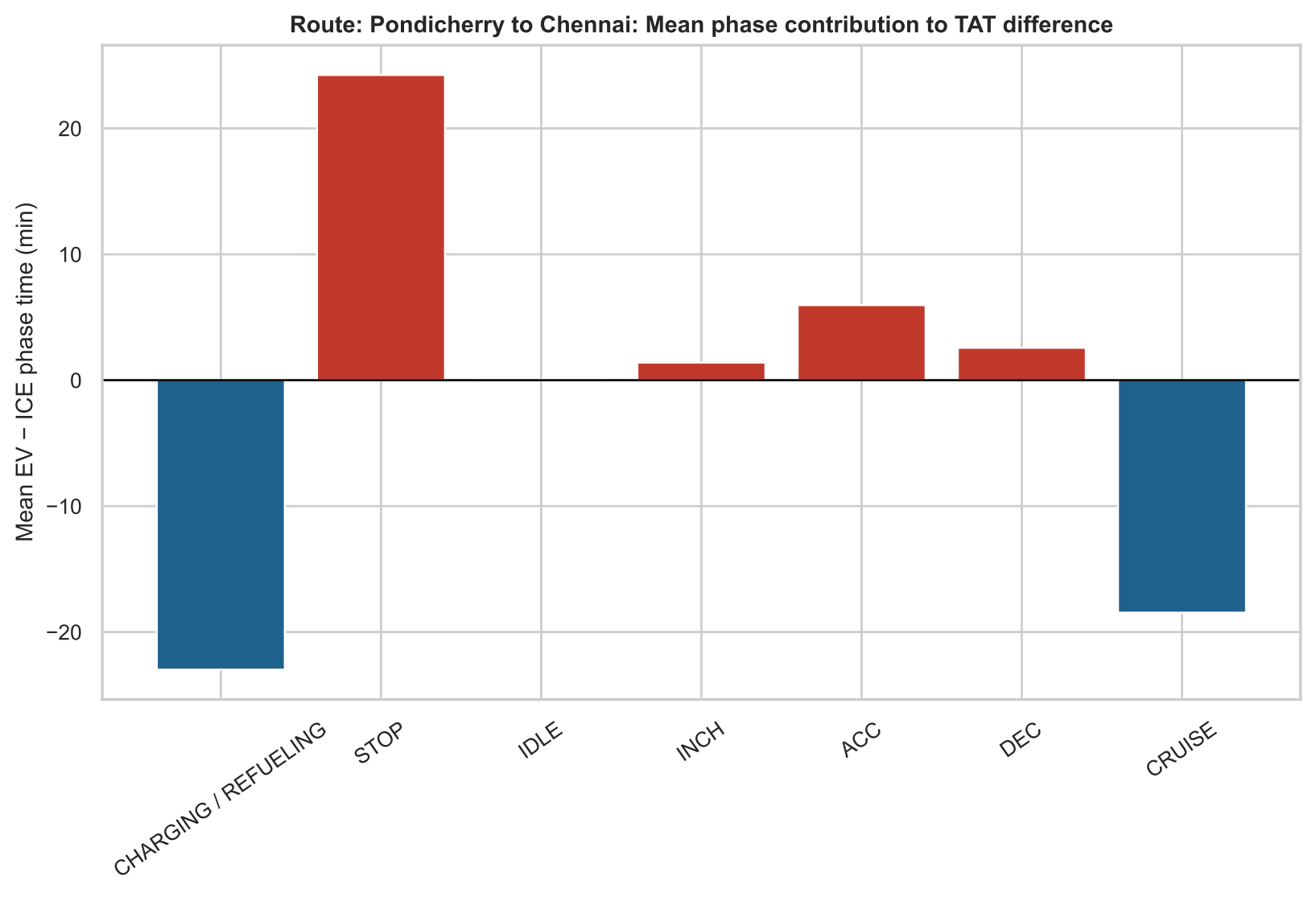
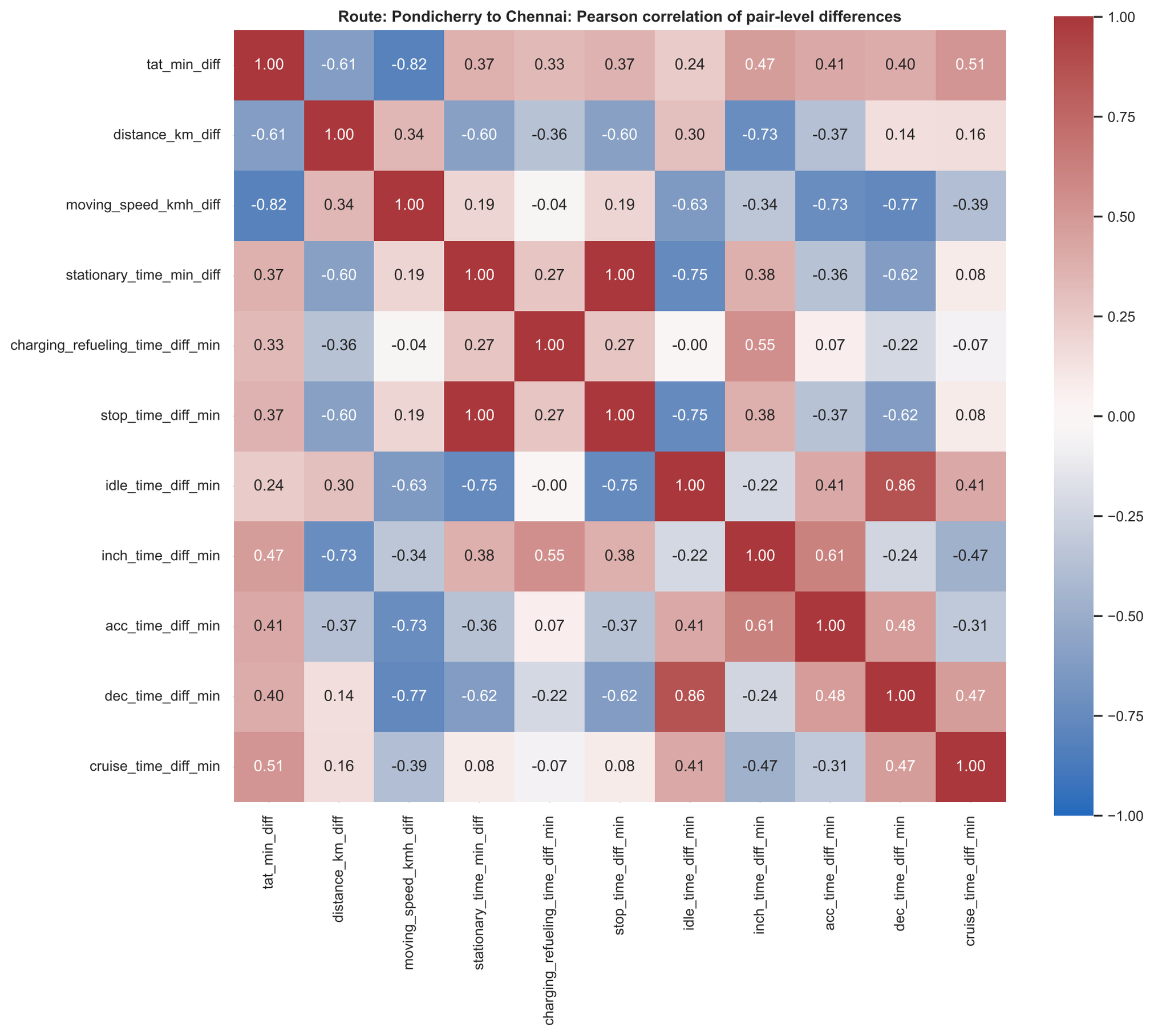

Route: Pondicherry to Chennai: Ranked pair-level TAT difference
PAIR_009
PAIR_008
PAIR_006
PAIR_014
PAIR_018
PAIR_015
PAIR_007
PAIR_010
PAIR_016
−20
−15
−10
−5
0
5
10
EV TAT − ICE TAT (min)

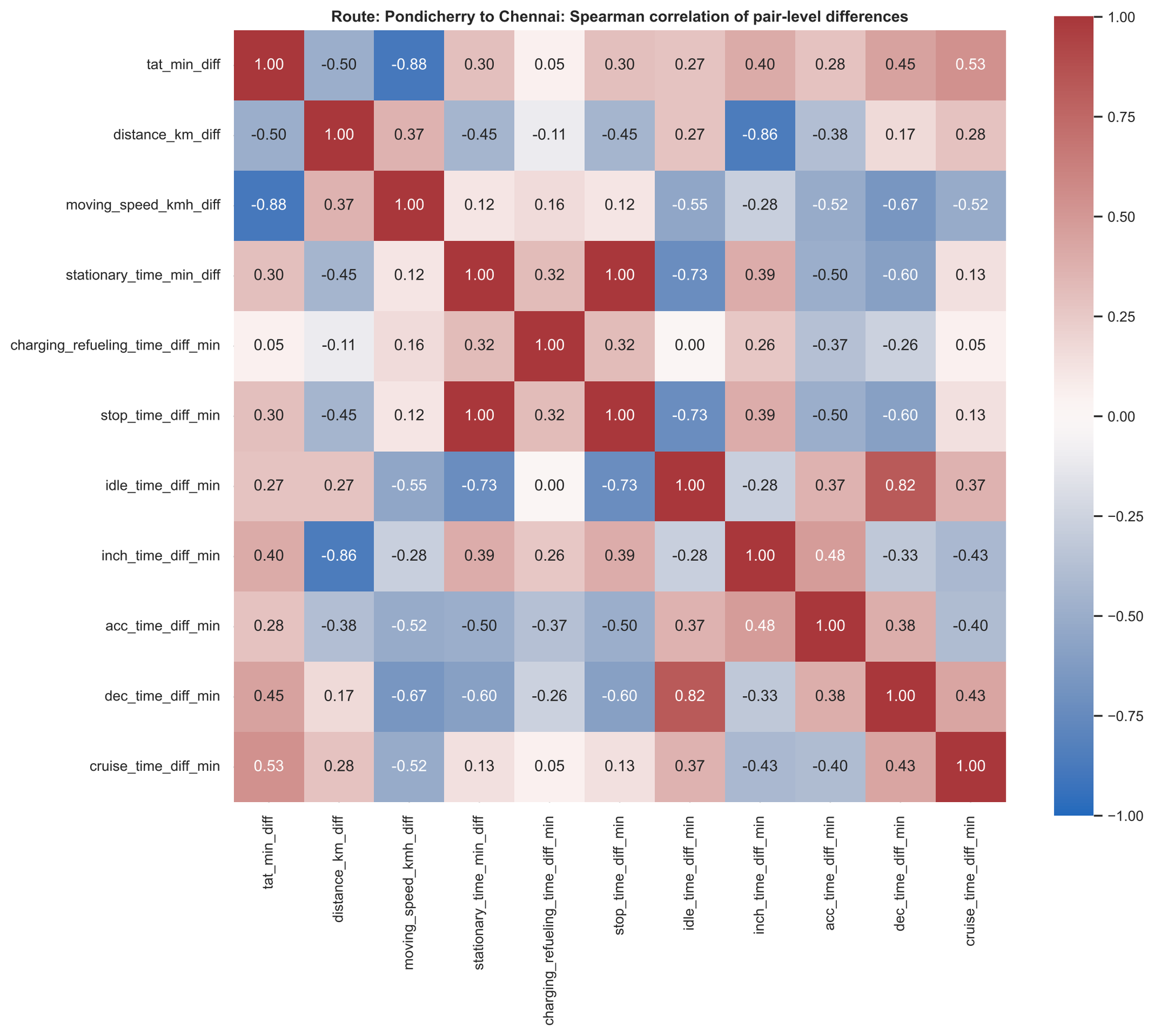

Route: Pondicherry to Chennai: Distribution of TAT differences
Count
3.0
2.5
2.0
1.5
1.0
0.5
0.0
−20
−15
−10
−5
0
5
10
EV TAT − ICE TAT (min)

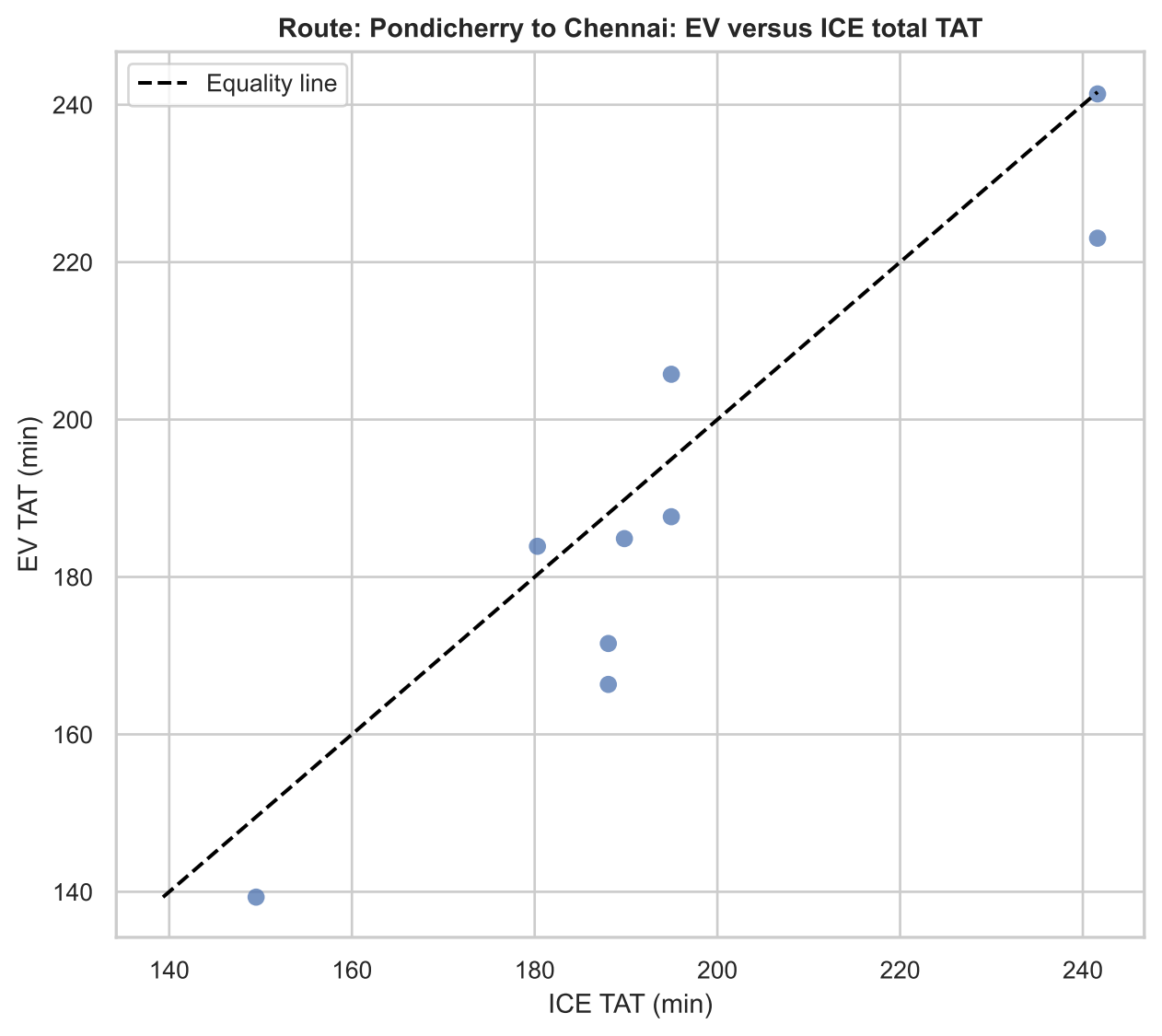

Route: Pondicherry to Chennai: EV and ICE TAT distribution
Total TAT (min)
240
220
200
180
160
140
EV
ICE

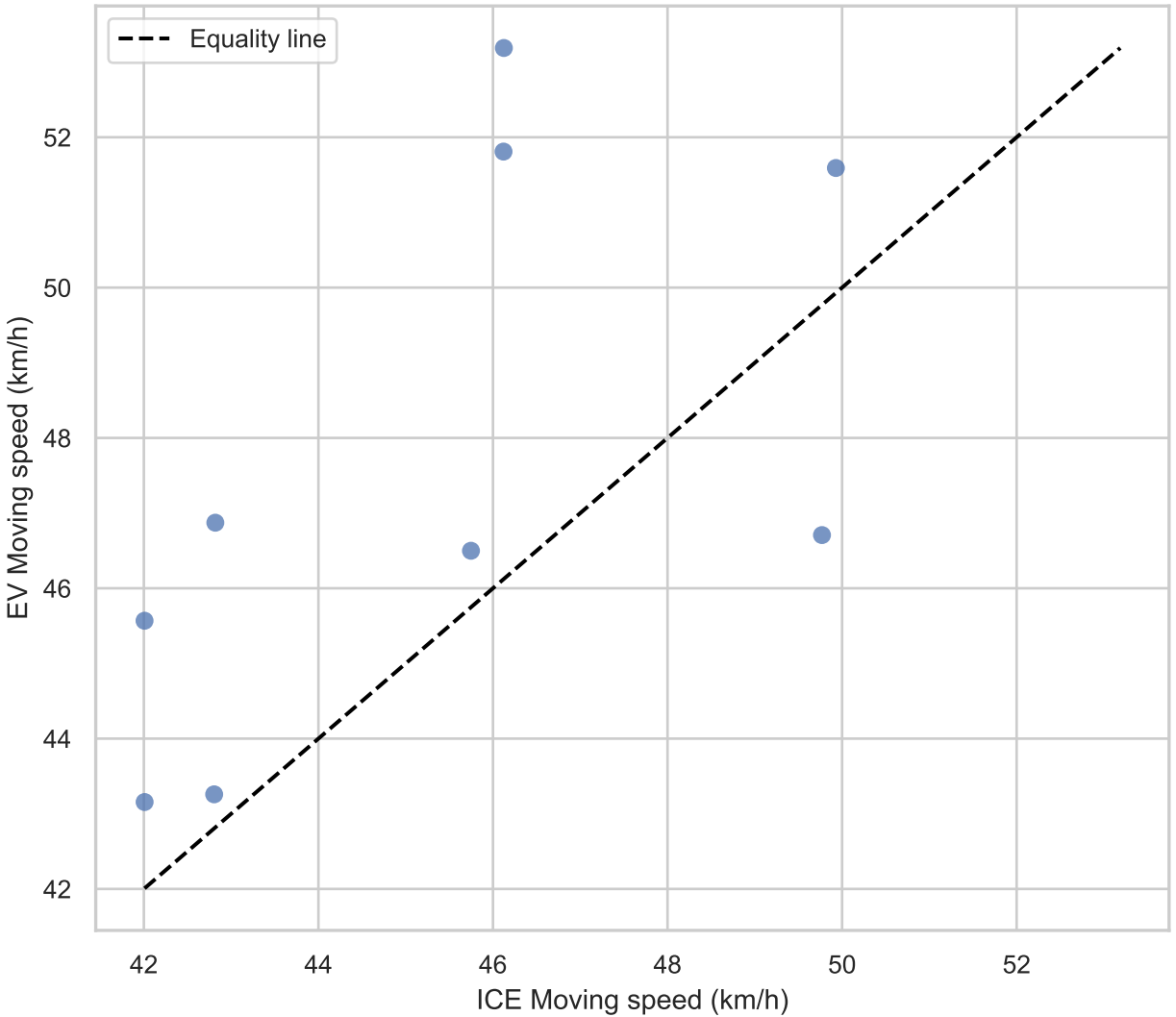

Route: Pondicherry to Chennai: Pair-by-phase time difference
pair_id
PAIR_016
PAIR_010
PAIR_007
PAIR_015
PAIR_018
PAIR_014
PAIR_006
PAIR_008
PAIR_009
CHARGING / REFUELING
STOP
IDLE
INCH
ACC
DEC
CRUISE
EV − ICE phase time (min)
20
10
0
−10
−20

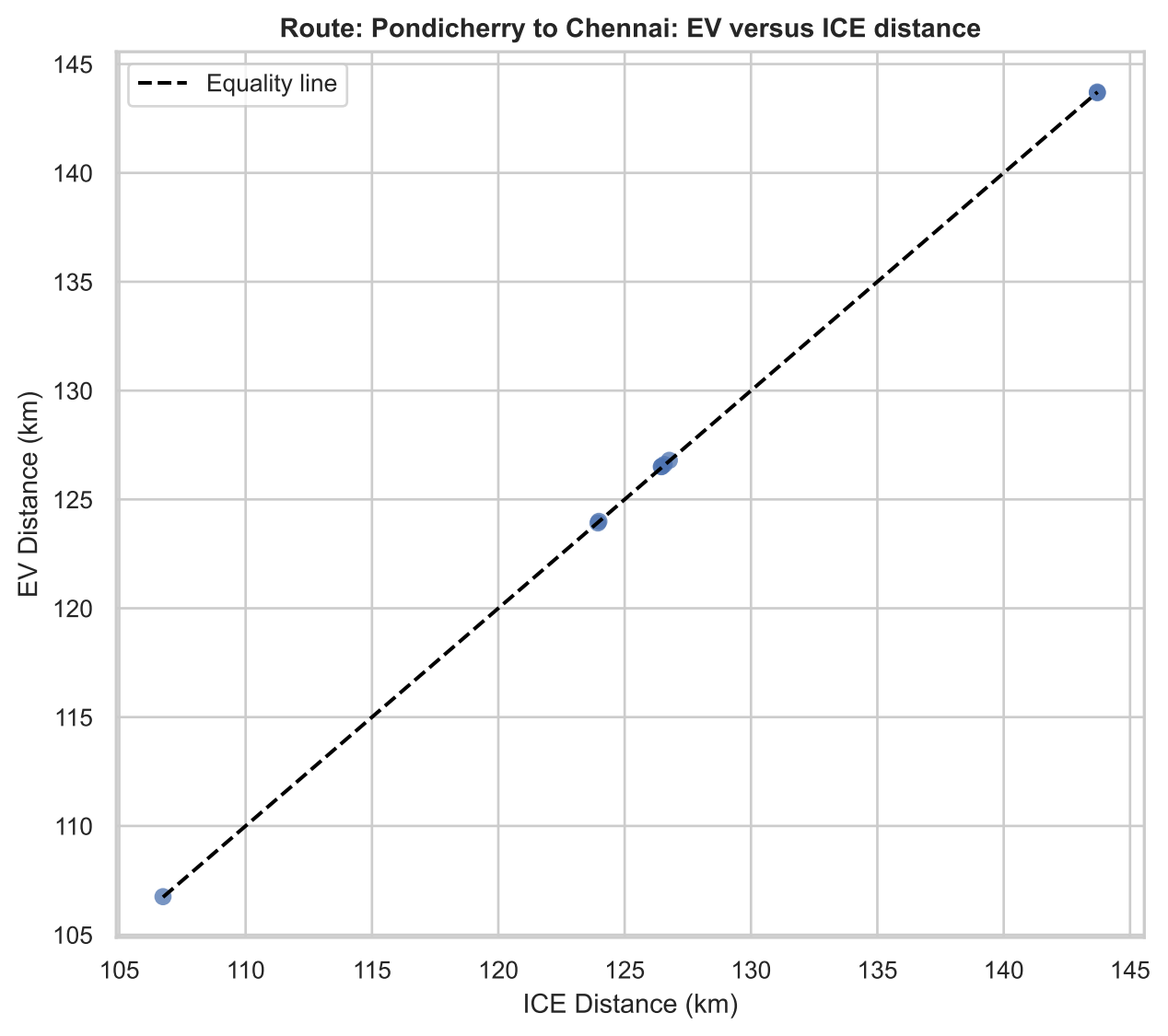

Route: Pondicherry to Chennai: Mean phase contribution to TAT difference
Mean EV – ICE phase time (min)
20
10
0
−10
−20
CHARGING / REFUELING
STOP
IDLE
INCH
ACC
DEC
CRUISE

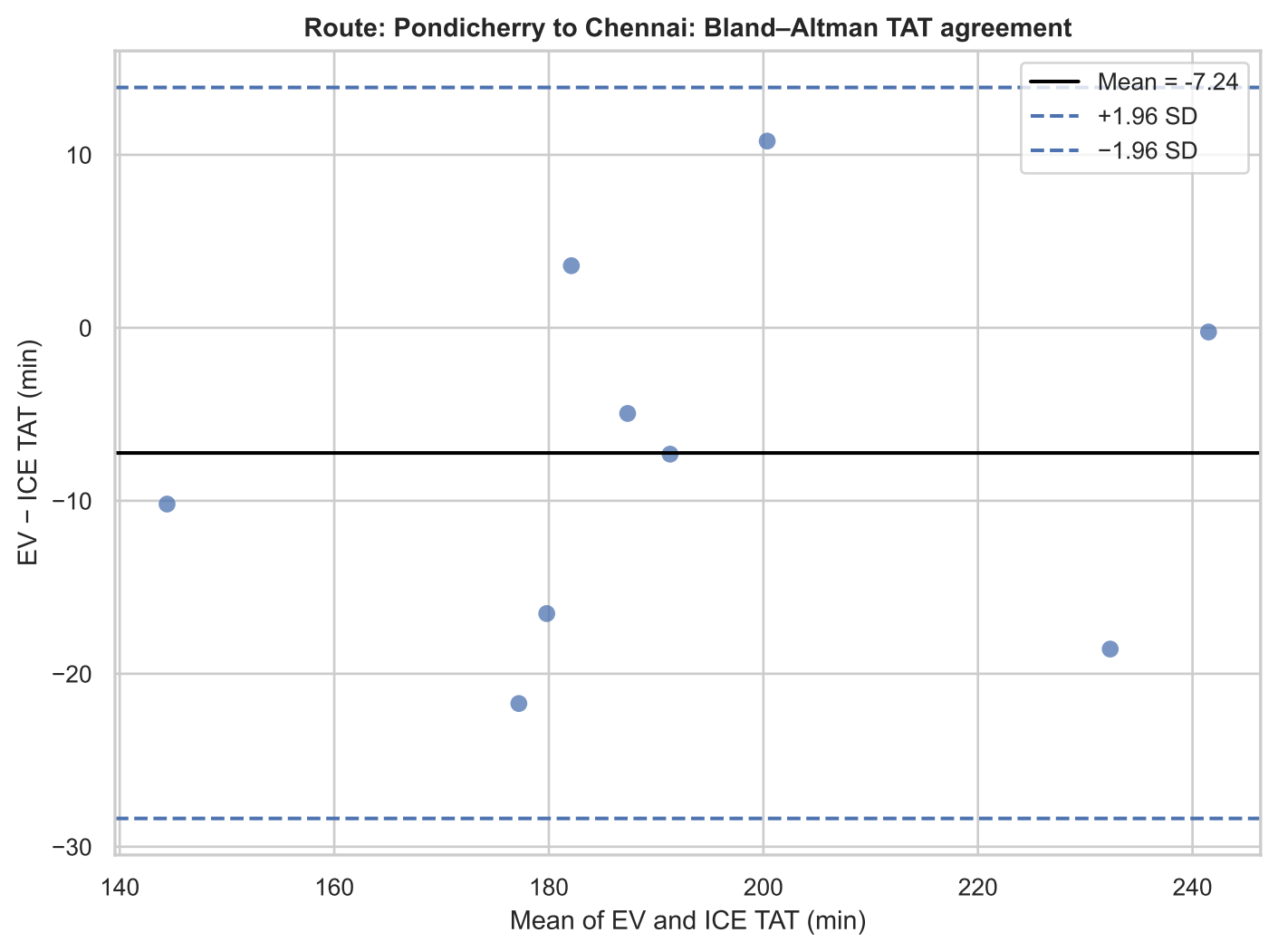

Route: Pondicherry to Chennai: Pearson correlation of pair-level differences
tat_min_diff 1.00 -0.61 -0.82 0.37 0.33 0.37 0.24 0.47 0.41 0.40 0.51
distance_km_diff -0.61 1.00 0.34 -0.60 -0.36 -0.60 0.30 -0.73 -0.37 0.14 0.16
moving_speed_kmh_diff -0.82 0.34 1.00 0.19 -0.04 0.19 -0.63 -0.34 -0.73 -0.77 -0.39
stationary_time_min_diff 0.37 -0.60 0.19 1.00 0.27 1.00 -0.75 0.38 -0.36 -0.62 0.08
charging_refueling_time_diff_min 0.33 -0.36 -0.04 0.27 1.00 0.27 -0.00 0.55 0.07 -0.22 -0.07
stop_time_diff_min 0.37 -0.60 0.19 1.00 0.27 1.00 -0.75 0.38 -0.37 -0.62 0.08
idle_time_diff_min 0.24 0.30 -0.63 -0.75 -0.00 -0.75 1.00 -0.22 0.41 0.86 0.41
inch_time_diff_min 0.47 -0.73 -0.34 0.38 0.55 0.38 -0.22 1.00 0.61 -0.24 -0.47
acc_time_diff_min 0.41 -0.37 -0.73 -0.36 0.07 -0.37 0.41 0.61 1.00 0.48 -0.31
dec_time_diff_min 0.40 0.14 -0.77 -0.62 -0.22 -0.62 0.86 -0.24 0.48 1.00 0.47
cruise_time_diff_min 0.51 0.16 -0.39 0.08 -0.07 0.08 0.41 -0.47 -0.31 0.47 1.00
tat_min_diff
distance_km_diff
moving_speed_kmh_diff
stationary_time_min_diff
charging_refueling_time_diff_min
stop_time_diff_min
idle_time_diff_min
inch_time_diff_min
acc_time_diff_min
dec_time_diff_min
cruise_time_diff_min
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00

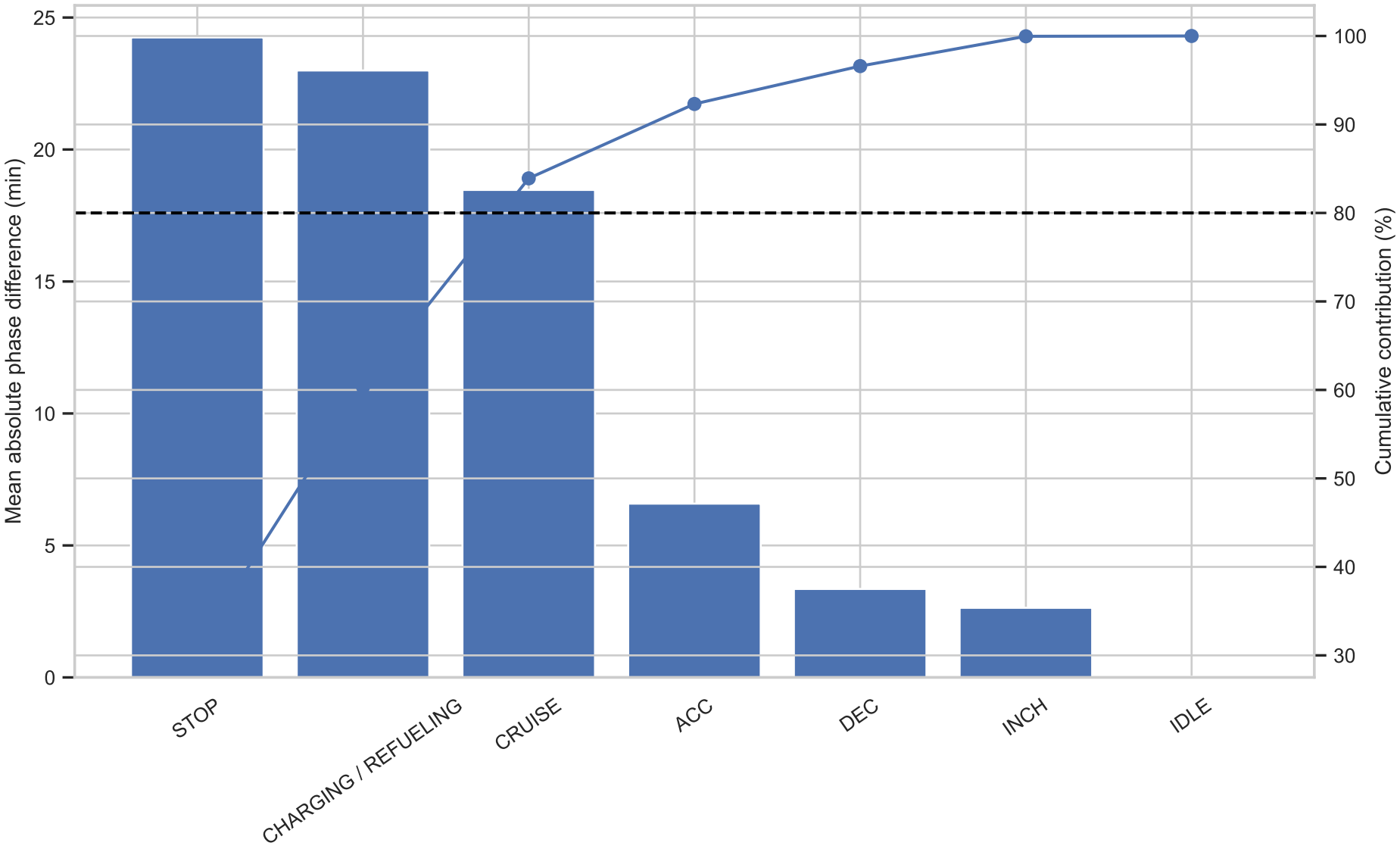

Route: Pondicherry to Chennai: Spearman correlation of pair-level differences
tat_min_diff 1.00 -0.50 -0.88 0.30 0.05 0.30 0.27 0.40 0.28 0.45 0.53
distance_km_diff -0.50 1.00 0.37 -0.45 -0.11 -0.45 0.27 -0.86 -0.38 0.17 0.28
moving_speed_kmh_diff -0.88 0.37 1.00 0.12 0.16 0.12 -0.55 -0.28 -0.52 -0.67 -0.52
stationary_time_min_diff 0.30 -0.45 0.12 1.00 0.32 1.00 -0.73 0.39 -0.50 -0.60 0.13
charging_refueling_time_diff_min 0.05 -0.11 0.16 0.32 1.00 0.32 0.00 0.26 -0.37 -0.26 0.05
stop_time_diff_min 0.30 -0.45 0.12 1.00 0.32 1.00 -0.73 0.39 -0.50 -0.60 0.13
idle_time_diff_min 0.27 0.27 -0.55 -0.73 0.00 -0.73 1.00 -0.28 0.37 0.82 0.37
inch_time_diff_min 0.40 -0.86 -0.28 0.39 0.26 0.39 -0.28 1.00 0.48 -0.33 -0.43
acc_time_diff_min 0.28 -0.38 -0.52 -0.50 -0.37 -0.50 0.37 0.48 1.00 0.38 -0.40
dec_time_diff_min 0.45 0.17 -0.67 -0.60 -0.26 -0.60 0.82 -0.33 0.38 1.00 0.43
cruise_time_diff_min 0.53 0.28 -0.52 0.13 0.05 0.13 0.37 -0.43 -0.40 0.43 1.00
tat_min_diff
distance_km_diff
moving_speed_kmh_diff
stationary_time_min_diff
charging_refueling_time_diff_min
stop_time_diff_min
idle_time_diff_min
inch_time_diff_min
acc_time_diff_min
dec_time_diff_min
cruise_time_diff_min
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00

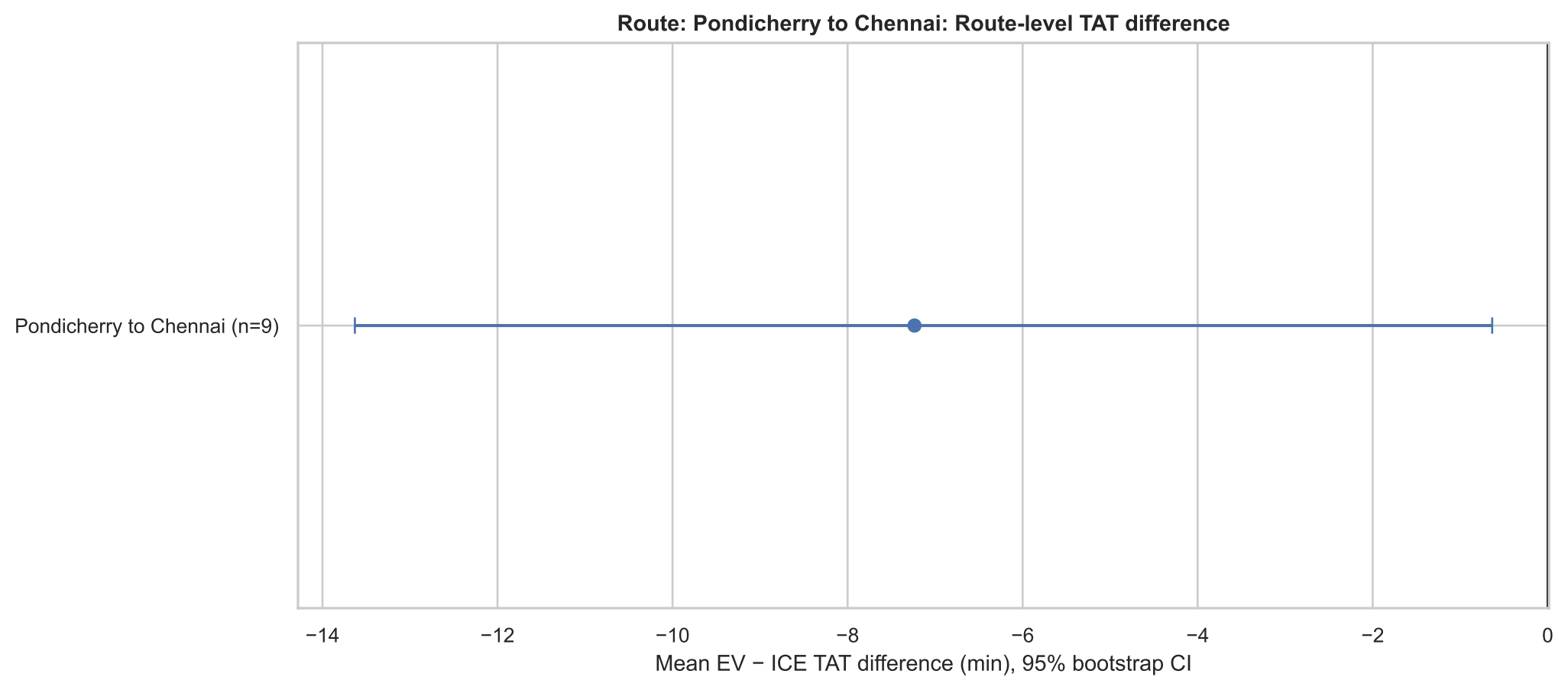

Route: Pondicherry to Chennai: EV versus ICE total TAT
Equality line
EV TAT (min)
ICE TAT (min)
140
160
180
200
220
240

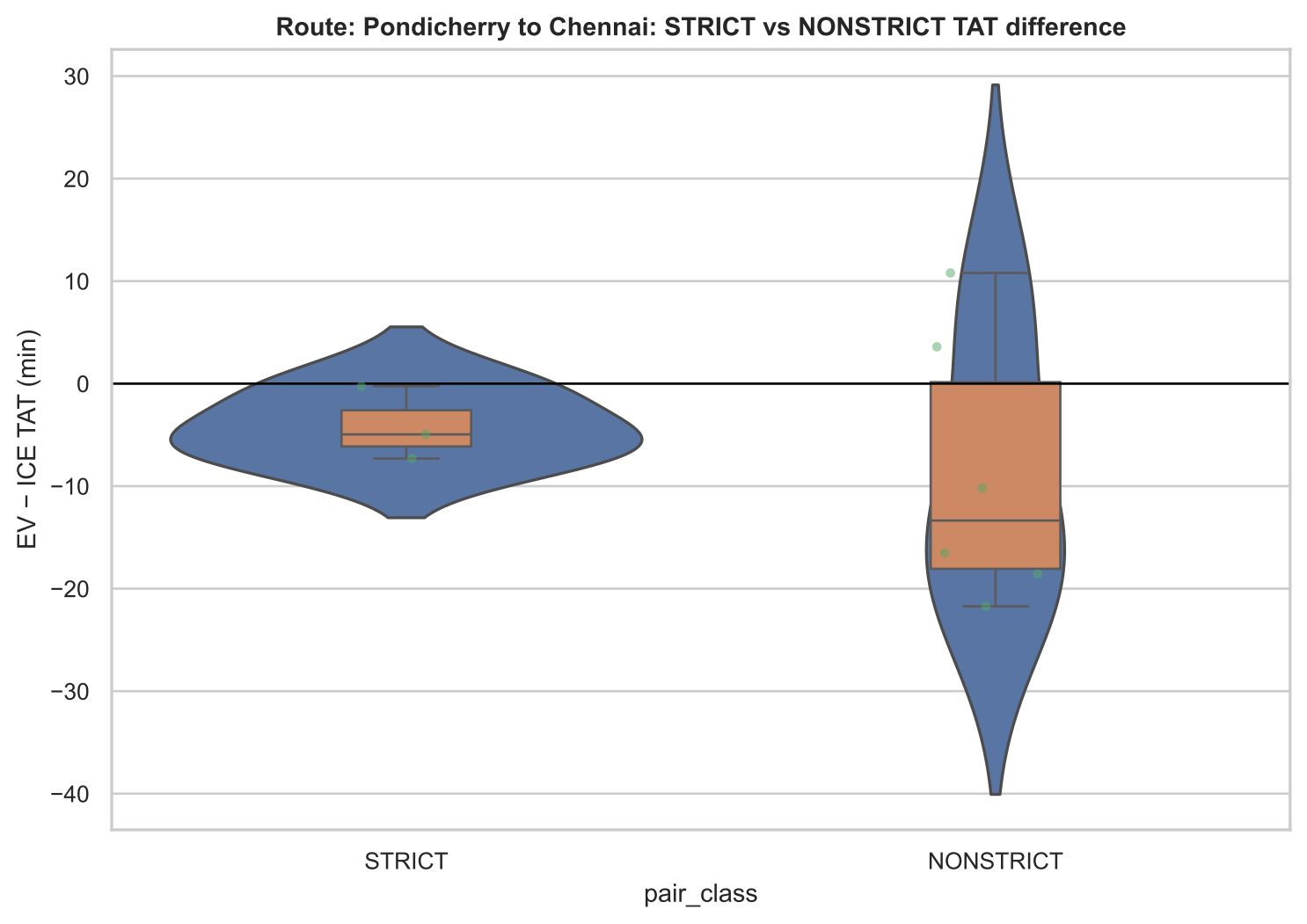

Route: Pondicherry to Chennai: EV versus ICE moving speed
Equality line
EV Moving speed (km/h)
ICE Moving speed (km/h)
42
44
46
48
50
52

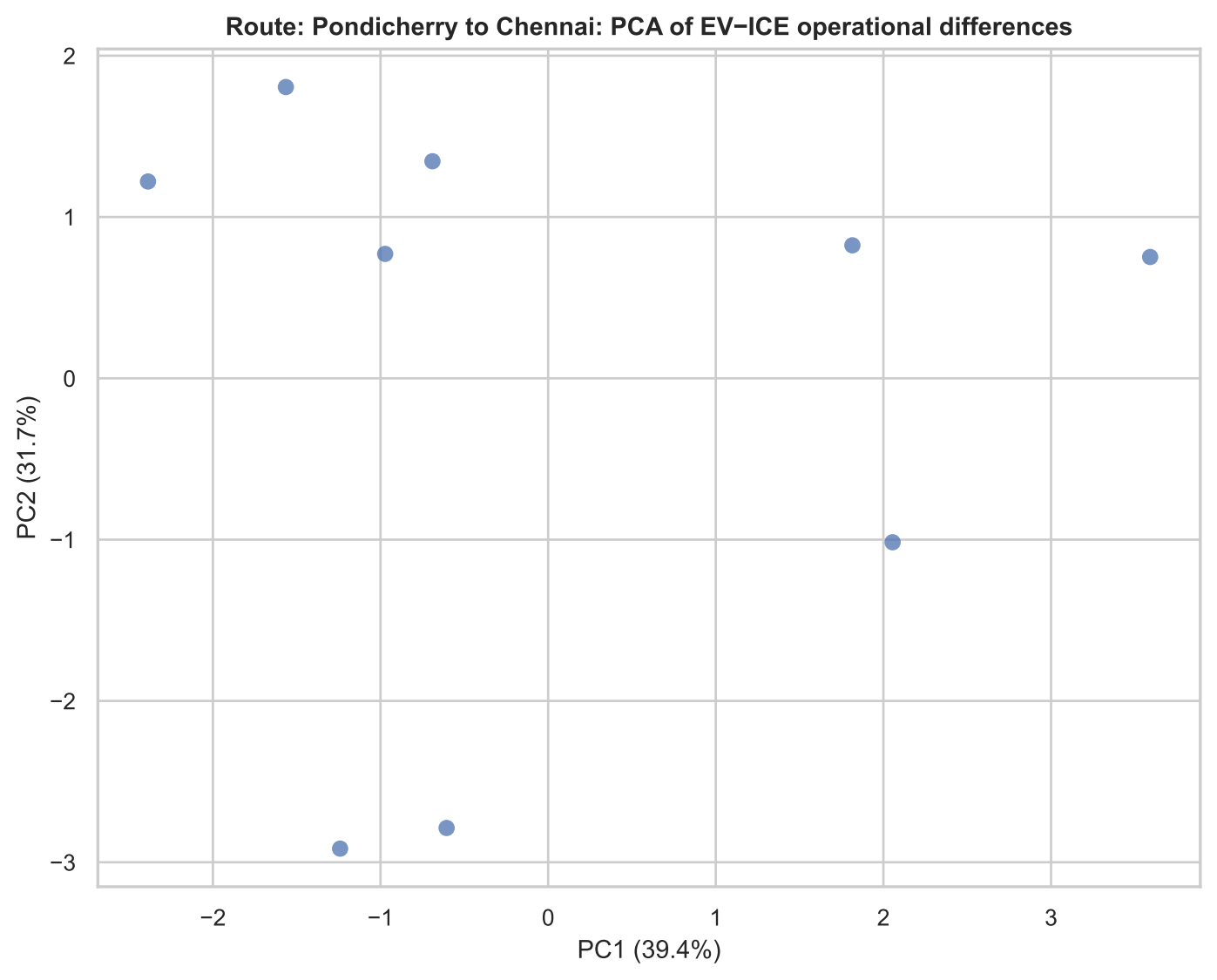

Route: Pondicherry to Chennai: EV versus ICE distance
Equality line
EV Distance (km)
ICE Distance (km)
105
110
115
120
125
130
135
140
145

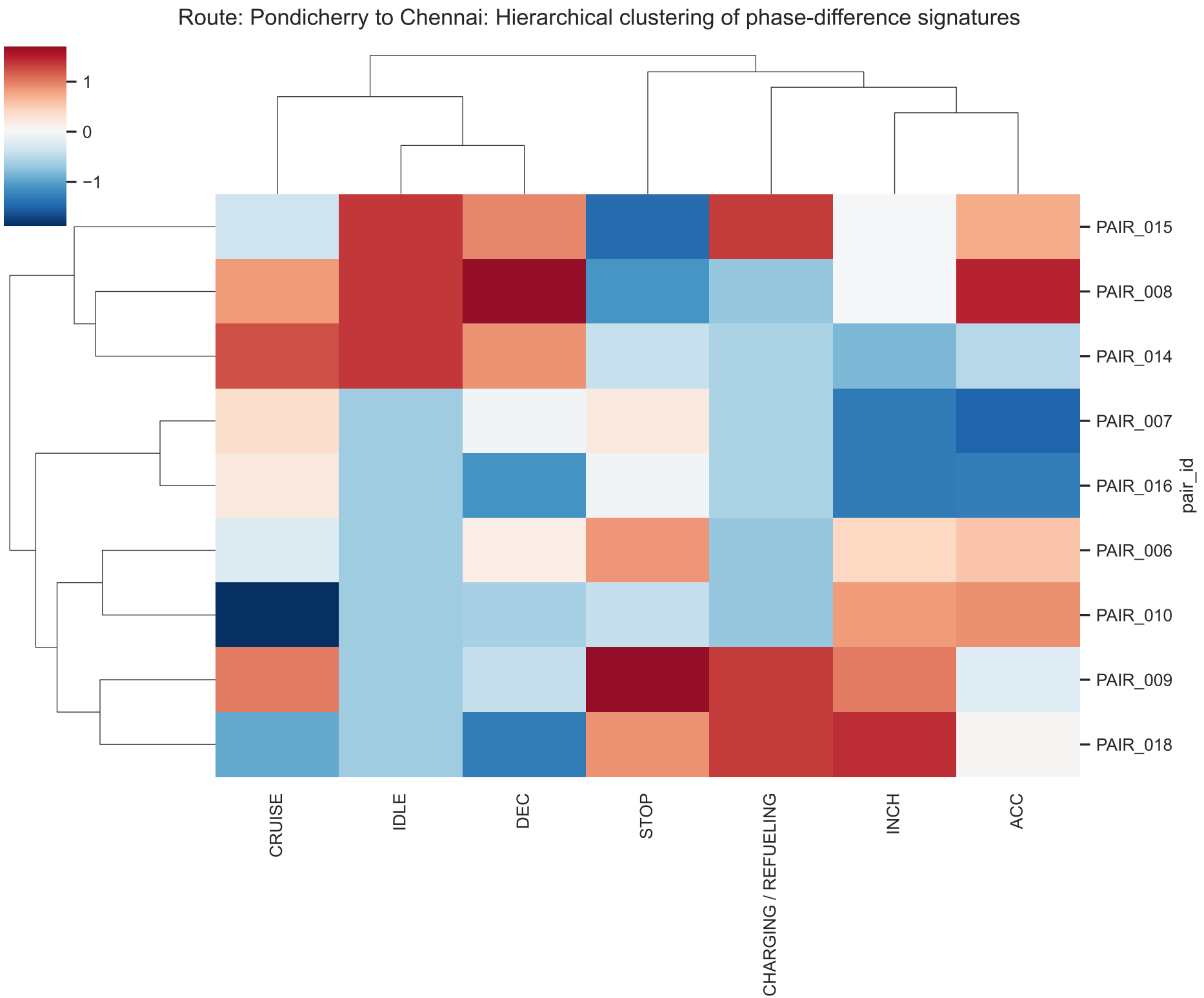

Route: Pondicherry to Chennai: Bland–Altman TAT agreement
Mean = -7.24
+1.96 SD
−1.96 SD
10
0
−10
−20
−30
EV – ICE TAT (min)
140
160
180
200
220
240
Mean of EV and ICE TAT (min)


Route: Pondicherry to Chennai: Pareto analysis of phase effects
Mean absolute phase difference (min)
Cumulative contribution (%)
0
5
10
15
20
25
30
40
50
60
70
80
90
100
STOP
CHARGING / REFUELING
CRUISE
ACC
DEC
INCH
IDLE


Route: Pondicherry to Chennai: Route-level TAT difference
Pondicherry to Chennai (n=9)
−14
−12
−10
−8
−6
−4
−2
0
Mean EV − ICE TAT difference (min), 95% bootstrap CI


Route: Pondicherry to Chennai: STRICT vs NONSTRICT TAT difference
EV – ICE TAT (min)
30
20
10
0
−10
−20
−30
−40
STRICT
NONSTRICT
pair_class


Route: Pondicherry to Chennai: PCA of EV−ICE operational differences
PC2 (31.7%)
PC1 (39.4%)
2
1
0
−1
−2
−3
−2
−1
0
1
2
3


Route: Pondicherry to Chennai: Hierarchical clustering of phase-difference signatures
1
0
−1
PAIR_015
PAIR_008
PAIR_014
PAIR_007
PAIR_016
PAIR_006
PAIR_010
PAIR_009
PAIR_018
pair_id
CRUISE
IDLE
DEC
STOP
CHARGING / REFUELING
INCH
ACC